

Spatio-Temporal Analysis of US Road Construction and Closures (2016–2021)

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Abstract—This paper presents a comprehensive spatio-temporal analysis of the US Road Construction and Closures dataset (2016–2021), comprising 6.17 million raw records across all 50 US states. After preprocessing, 1,644,242 high-quality records are retained. We engineer 19 spatial and temporal features including cyclical time encodings, grid-based density proxies, POI aggregate counts, and geospatial attributes. For the temporal dimension, we apply STL decomposition, ADF stationarity testing, ARIMA, SARIMA, ETS, Prophet, and LSTM forecasting, alongside severity classification and duration regression. For the spatial dimension, we deploy Moran’s I, Geary’s C, LISA, kernel density estimation, K-Means and DBSCAN clustering, spatial classification, geographically weighted regression, and variogram analysis. Data mining encompasses spatial colocation (Participation Index), temporal motif discovery, Apriori association rule mining, Isolation Forest and LOF anomaly detection, and Getis-Ord G^* hotspot analysis. Joint inference integrates spatial and temporal features in XGBoost, Random Forest, and LSTM models. Key findings reveal Florida, New York, and California as persistent hotspots, strong summer seasonality, a COVID-19 disruption in 2020, and that joint spatio-temporal features outperform unimodal feature sets for both classification and regression tasks. Code repository: <https://github.com/jitin1975/us-road-closures-spatiotemporal-analysis>.

Index Terms—spatio-temporal analysis, road closures, time series, spatial autocorrelation, colocation mining, ARIMA, SARIMA, Prophet, LSTM, XGBoost, hotspot analysis, association rule mining

I. INTRODUCTION

Road construction and closures impose significant costs on transportation networks, economic productivity, and public safety. Understanding *when* and *where* closures occur—and predicting their duration—is essential for infrastructure planners and emergency responders. Spatio-temporal analysis offers a principled methodology for uncovering patterns that are invisible to purely spatial or purely temporal analyses taken in isolation.

This work analyses the US Road Construction and Closures dataset [1] compiled from state Department of Transportation feeds, incident reporting systems, and weather APIs. The dataset covers approximately five years (February 2016 to December 2021) across all 50 US states, and provides rich contextual attributes including weather conditions, point-of-interest (POI) proximity flags, and severity ratings.

Our primary research questions are: (i) Are road closure events spatially clustered, and which geographic features ex-

plain that clustering? (ii) Do closure frequencies exhibit statistically significant temporal trends and seasonality? (iii) Can machine learning models predict closure severity or duration from spatial and temporal features alone and in combination? (iv) What colocation and association patterns exist among infrastructure POI types and temporal contexts? (v) Which spatial and temporal feature sets provide the greatest predictive lift when combined?

We address these questions through a four-part pipeline: EDA and feature engineering (Section II), temporal analysis and ML (Section III), spatial analysis and ML (Section IV), and data mining with joint inference (Section V). Results and discussion appear in Section VI, followed by conclusions in Section VII.

II. DATASET DESCRIPTION AND FEATURE ENGINEERING

A. Dataset Overview

The raw dataset contains 6,170,627 records and 47 attributes. Each record represents a single road construction or closure event with a start and end timestamp, GPS coordinates (start/end latitude and longitude), severity rating (1–4), textual description, street-level address, meteorological readings at the time of the event, and 13 binary POI proximity flags (e.g., Junction, Traffic_Signal, Crossing, Railway). Table I summarises the dataset at a high level.

TABLE I
DATASET OVERVIEW: US ROAD CONSTRUCTION AND CLOSURES (2016–2021)

Metric	Value
Total Raw Records	6,170,627
Records after Cleaning	1,644,242
Date Range Start	2016-02-04
Date Range End	2021-12-31
Total States	50
Total Cities	12,999
Missing Values (%)	1.92%

B. Preprocessing

Preprocessing involves: (i) parsing `Start_Time` and `End_Time` as timezone-aware datetimes; (ii) dropping rows with missing latitude, longitude, or start time; (iii) computing event duration $\Delta t = \text{End_Time} - \text{Start_Time}$ in hours,

capped at 720 h (30 days) to remove manifest data errors; and (iv) filtering records to the US mainland bounding box (24.5–49.5N, 125–66W). After these steps, 1,644,242 records are retained, representing 26.6% of raw data, with 85 total columns after feature engineering. The data’s overall missing-value rate is only 1.92%, confirming high completeness for numerical fields.

Table II presents descriptive statistics. The mean event duration is 32.9 h with a heavy right tail (max 720 h); the median is only 6 h, indicating that most events are short-lived while a small fraction of long-running closures inflate the mean. Distance affected has a median of 0.33 miles.

TABLE II
DESCRIPTIVE STATISTICS OF KEY NUMERICAL FEATURES (AFTER PREPROCESSING)

Feature	Mean	Median	Std	Max
Duration (hrs)	32.88	6.00	80.59	720.0
Distance (mi)	1.23	0.33	3.83	574.9
Temperature (°F)	63.19	66.00	17.97	196.0
Humidity (%)	63.30	65.00	22.75	100.0
Wind Speed (mph)	7.30	7.00	5.58	983.0
Visibility (mi)	9.44	10.00	26.95	34007.0

C. Exploratory Data Analysis

Figure 1 shows temporal distributions of events across hour-of-day, day-of-week, month, and year. Event frequency peaks between 07:00–09:00 and 16:00–18:00, aligning with morning and evening rush hours. A strong summer seasonality is evident, with June–August recording the highest monthly counts. Annual volume grew steadily from 2016 to 2019, dipped noticeably in 2020 (consistent with reduced road activity during the COVID-19 pandemic), and recovered in 2021.

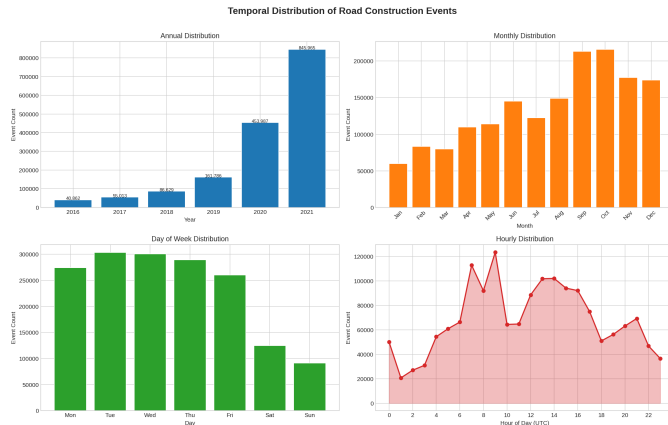


Fig. 1. Temporal distributions of road closure events: (a) hour-of-day showing rush-hour peaks, (b) day-of-week with weekday dominance, (c) month with summer peaks, and (d) year showing COVID-19 disruption in 2020.

Figure 2 illustrates the closure severity distribution. Severity 2 events dominate (approximately 70%), while Severity 4

events are rare (<2%), suggesting most closures are moderate inconveniences rather than complete road blockages.

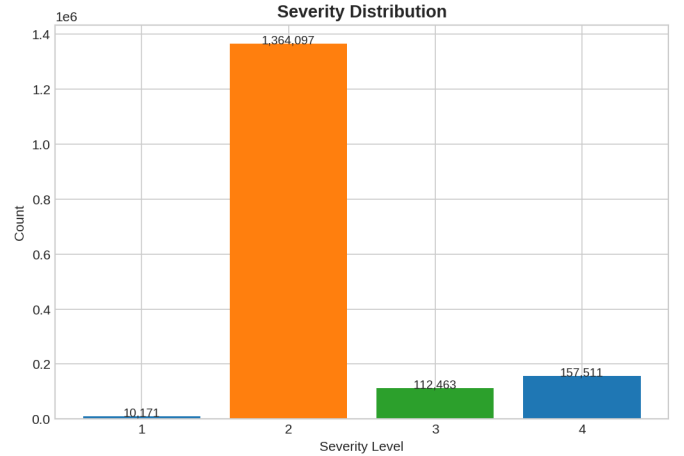


Fig. 2. Distribution of closure severity levels (1–4) and closure type frequencies. Severity 2 accounts for the majority of events; construction-type closures are most common.

State-level analysis (Table III) shows Florida (FL, 207,708 events), New York (NY, 157,261), and California (CA, 156,385) as the top three states, collectively accounting for 32% of all records. Notably, New Jersey has the highest average duration (41.2 h), while Florida has one of the lowest (21.6 h) despite the highest event count, suggesting shorter, more frequent maintenance activities.

TABLE III
STATE-LEVEL CONSTRUCTION EVENT STATISTICS (TOP 15 BY COUNT)

State	Event Count	Avg Duration (h)	Median Duration (h)
FL	207,708	21.62	4.15
NY	157,261	27.89	5.64
CA	156,385	29.05	7.75
PA	129,302	37.53	6.75
AZ	99,682	32.87	2.96
TX	91,265	30.07	4.45
IL	72,020	29.08	4.00
CO	68,969	31.48	7.52
NJ	41,030	41.24	9.39
IN	37,143	37.58	4.16
OH	36,780	48.41	6.00
VA	36,607	31.36	8.00
GA	36,123	38.89	8.00
WA	35,810	32.71	6.04
OR	34,362	23.15	4.00

D. Feature Engineering

We engineer 19 features across four categories as summarised in Table IV. Principal Component Analysis (PCA) on the engineered feature set (Figure 3) shows that the first two components explain approximately 38% of total variance, with temporal cyclic features clustering distinctly from spatial density features.

Temporal features: Cyclical encodings of hour, month, day-of-week, and week-of-year use sine/cosine transformations (e.g., $h_{\sin} = \sin(2\pi \cdot \text{hour}/24)$) to preserve periodicity in Euclidean distance-based models. Binary flags `IsWeekend`, `IsRushHour` (06:00–09:00 and 16:00–19:00), and `IsNight` (20:00–06:00) capture behavioural contexts. Meteorological Season and four-level `TimeOfDay` categories are also derived.

Spatial features: Each event is assigned to a $1^\circ \times 1^\circ$ and $0.5^\circ \times 0.5^\circ$ lat/lon grid cell, with per-cell event density computed as an intensity proxy. Euclidean distance from the US geographic centroid (39.5°N, 98.4°W) captures centre-periphery gradients. Coast proximity (East/West/Interior) and an urban/rural proxy derived from grid density quartile rank are included. The sum of all 13 binary POI columns yields `POI_Count`.

Weather features: Temperature is binned into Cold ($<40^\circ\text{F}$), Moderate ($40\text{--}80^\circ\text{F}$), and Hot ($>80^\circ\text{F}$). Binary flags `HasPrecipitation` and `LowVisibility` (visibility <3 mi) encode adverse weather conditions.

TABLE IV
ENGINEERED FEATURES FOR SPATIO-TEMPORAL ANALYSIS

Feature	Description	Category
Hour_sin/cos	Cyclical hour encoding	Temporal
Month_sin/cos	Cyclical month encoding	Temporal
DOW_sin/cos	Cyclical day-of-week encoding	Temporal
Week_sin/cos	Cyclical week-of-year encoding	Temporal
IsWeekend	Binary weekend flag	Temporal
IsRushHour	Peak traffic hours flag	Temporal
IsNight	Nighttime flag (20:00–06:00)	Temporal
Season	Meteorological season	Temporal
TimeOfDay	Time-of-day category	Temporal
StateMonthCount	State-month event density	Spatio-Temp.
GridCell/Density	1° spatial grid & density	Spatial
HexCell/Density	0.5° hex grid & density	Spatial
DistFromCenter	Distance from US centroid	Spatial
POI_Count	Aggregate nearby POI count	Spatial
CoastProximity	Proximity to coast	Spatial
IsUrban	Urban/rural classification	Spatial
LowVisibility	Visibility below 3 mi flag	Weather
HasPrecipitation	Precipitation presence flag	Weather
TempCategory	Temperature discretization	Weather

E. Feature Engineering Rationale

Spatial characteristics used: The dataset provides GPS coordinates (latitude/longitude) for each event, enabling derivation of grid-cell assignment ($1^\circ \times 1^\circ$ and $0.5^\circ \times 0.5^\circ$), per-cell event density as an intensity proxy, and Euclidean distance from the US geographic centroid to capture centre-periphery gradients. The 13 binary POI proximity flags (Junction, Traffic_Signal, Crossing, Railway, etc.) are aggregated into a scalar

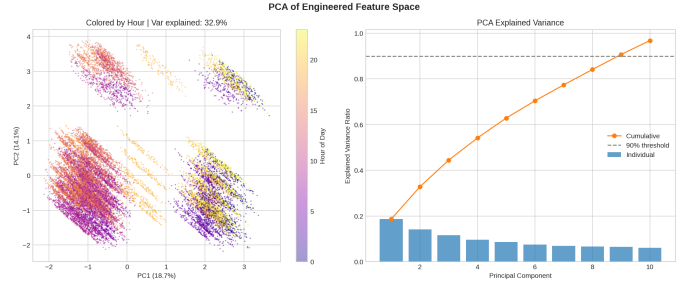


Fig. 3. PCA biplot of engineered features. The first two principal components explain $\approx 38\%$ of variance. Temporal cyclic features (hour, month) cluster separately from spatial density features (GridDensity, StateMonthCount).

`POI_Count` and also used individually in colocation and association mining. Coast proximity (East/West/Interior) and an urban/rural proxy derived from grid-density quartile rank capture macro-scale spatial structure. These spatial features are motivated by the observation that event density, infrastructure type, and urban context directly influence where closures concentrate and how long they last.

Temporal characteristics used: The `Start_Time` field supplies all temporal signal. Cyclical sine/cosine encodings of hour-of-day, day-of-week, month, and week-of-year preserve periodicity for distance-based models. Binary flags `IsWeekend`, `IsRushHour` (06:00–09:00 and 16:00–19:00), and `IsNight` (20:00–06:00) encode behavioural contexts aligned with observed rush-hour peaks in Figure 1. Meteorological season and a four-level `TimeOfDay` category are derived from the timestamp. The hybrid feature `StateMonthCount` (state-month event density) bridges temporal and spatial domains. These features are motivated by the strong 12-month seasonality and rush-hour concentration revealed in the EDA (Section II-C).

Weather characteristics used: Meteorological readings at event time (Temperature, Humidity, Wind_Speed, Visibility, Precipitation_Type) are discretised into binary flags `HasPrecipitation` and `LowVisibility` (visibility <3 mi), and `TempCategory` (Cold/Moderate/Hot). Weather features proved the weakest predictor set in joint inference ($R^2 = 0.033$ for regression alone), consistent with weather being largely subsumed by the spatial and temporal dimensions.

III. TEMPORAL ANALYSIS AND MACHINE LEARNING

A. Time Series Construction and Decomposition

Daily event counts form a time series of 2,158 days (February 2016–December 2021); monthly aggregation yields 71 monthly observations used for forecasting. Figure 4 shows the STL decomposition of the monthly count series into trend, seasonal, and residual components. The trend component confirms sustained growth from 2016 to 2019 followed by a sharp COVID-19 dip in 2020 and recovery through 2021. The seasonal component has a clear 12-month period peak-

ing in summer (June–August). Residuals are approximately stationary after trend and seasonality removal.

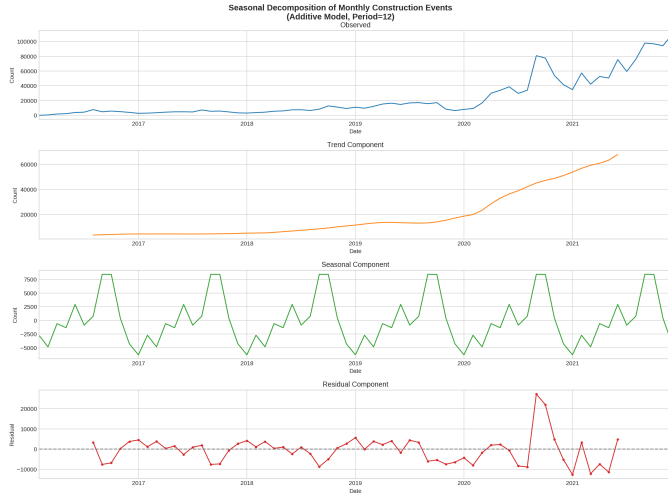


Fig. 4. STL decomposition of monthly road closure event counts (2016–2021) into trend (sustained growth then COVID dip), seasonal (12-month summer peak), and residual components.

The hour-of-day by day-of-week intensity heatmap (Figure 5) reveals the highest closure activity during weekday mornings (07:00–09:00) and afternoons (14:00–18:00), with weekends showing markedly lower and more uniformly distributed activity.

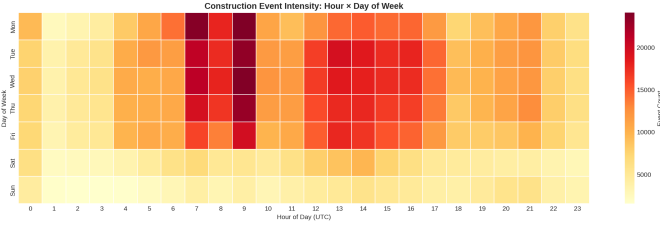


Fig. 5. Heat map of road closure event intensity by hour-of-day (rows) vs. day-of-week (columns). Peak activity clusters on weekday mornings and afternoons.

B. Stationarity Testing

The Augmented Dickey-Fuller (ADF) test [2] is applied to the original and differenced monthly series. As shown in Table V, the original series is non-stationary ($ADF = 3.22$, $p = 1.00$), but the first-differenced series achieves stationarity ($ADF = -6.36$, $p < 0.001$), satisfying the integration order $d = 1$ required for ARIMA-family models.

C. ARIMA Forecasting

ACF and PACF analysis (Figure 6) of the first-differenced monthly series reveals a significant lag-1 autocorrelation, supporting an MA(1) specification. A grid search over $(p, d, q) \in \{0, 1, 2\}^3$ with $d = 1$ fixed selects ARIMA(0,1,1) as optimal

TABLE V
AUGMENTED DICKEY-FULLER (ADF) STATIONARITY TEST RESULTS

Series	ADF Stat	p-value	Lags	Stationary?
Original (monthly)	3.2193	1.0000	10	No
1st Difference	−6.3581	0.0000	2	Yes
2nd Difference	−4.4429	0.0002	10	Yes

by AIC (Table VI). The model is trained on 56 months and evaluated on the final 15 months. ARIMA achieves $RMSE = 45,271$ and $MAPE = 77.2\%$ (Table VIII), reflecting difficulty in tracking the sharp COVID-19 anomaly.

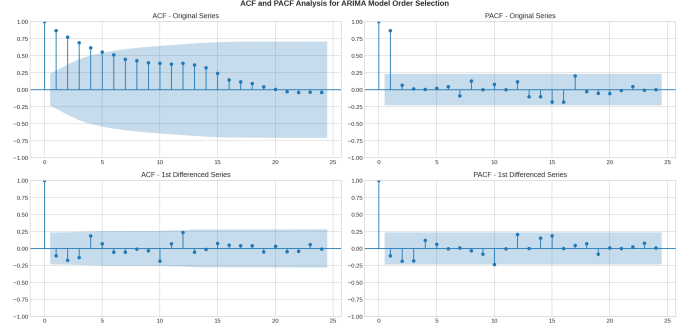


Fig. 6. ACF and PACF of the first-differenced monthly event count series. A significant lag-1 autocorrelation in the ACF with rapid decay confirms an MA(1) process, supporting ARIMA(0,1,1).

TABLE VI
ARIMA GRID SEARCH — TOP 5 ORDERS BY AIC

Order	AIC	BIC
(0, 1, 1)	1125.41	1129.43
(2, 1, 2)	1126.19	1136.23
(1, 1, 1)	1127.26	1133.28
(2, 1, 1)	1128.17	1136.20
(1, 1, 2)	1128.99	1137.02

D. Exponential Smoothing (ETS)

Four ETS variants are evaluated as shown in Table VII. Simple exponential smoothing (SES) achieves $RMSE = 26,009$ ($MAPE = 43.3\%$), outperforming both Holt linear ($RMSE = 87,566$) and additive Holt-Winters ($RMSE = 72,001$). Multiplicative Holt-Winters yields $RMSE = 29,883$. The poor performance of trend-augmented methods is attributable to the non-monotone COVID-19 disruption causing overshoot during trend extrapolation.

E. SARIMA

SARIMA(1,1,1)(1,1,0)₁₂ incorporates seasonal differencing and a seasonal AR(1) term, achieving $RMSE = 18,293$ and $MAPE = 14.2\%$ —the best result among classical time-series models. The seasonal AR component successfully captures the 12-month periodicity confirmed in STL decomposition.

TABLE VII
EXPONENTIAL SMOOTHING (ETS) MODEL COMPARISON (15-MONTH TEST SET)

Model	RMSE	MAPE (%)	R ²
Simple ES (α)	26,009	43.28	-0.319
Holt (trend)	87,566	135.73	-13.947
Holt-Winters (add.)	72,001	107.00	-9.106
Holt-Winters (mult.)	29,883	34.83	-0.741

F. Prophet Forecasting

Facebook Prophet [3] is configured with multiplicative seasonality and US holiday effects. As shown in Figure 7, it captures trend changepoints including the 2020 COVID-19 disruption more gracefully than ARIMA, yielding RMSE = 25,405 and MAPE = 20.6%.

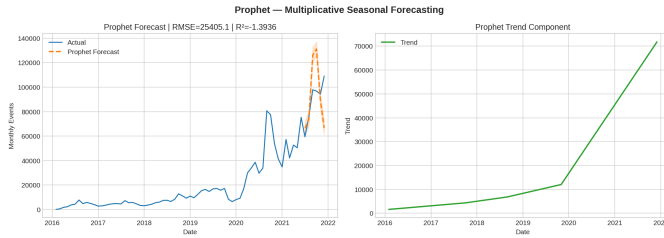


Fig. 7. Prophet forecast of monthly road closure event counts. Trend changepoints (vertical dashed lines) are automatically identified, including the 2020 COVID-19 disruption. Shaded region is the 80% uncertainty interval.

G. LSTM Temporal Forecasting

A two-layer LSTM (hidden size 64, dropout 0.2, look-back 6 months) is trained for 50 epochs on monthly counts. It achieves RMSE = 28,652 and MAPE = 29.3%. While capturing non-linear dynamics, the LSTM underperforms SARIMA on this 71-point monthly series due to limited training data for deep models.

Table VIII consolidates all temporal forecasting results.

TABLE VIII
TEMPORAL FORECASTING MODEL SUMMARY (15-MONTH TEST SET)

Model	RMSE	MAPE (%)	R ²
ARIMA(0,1,1)	45,271	77.24	-2.995
Simple ES	26,009	43.28	-0.319
Holt-Winters (mult.)	29,883	34.83	-0.741
Prophet (multiplicative)	25,405	20.62	-1.394
SARIMA(1,1,1)(1,1,0) ₁₂	18,293	14.19	-0.241
LSTM (look-back=6)	28,652	29.33	-2.045

H. Temporal Classification and Regression

Severity classification (predicting Severity 1–4 from temporal features) is evaluated on 100,000 events (80/20 split). Gradient Boosting achieves accuracy = 0.876 and macro-F1 = 0.475 (Table IX), while Logistic Regression reaches only accuracy = 0.543, confirming that non-linear methods are required for this task.

TABLE IX
TEMPORAL CLASSIFICATION MODELS: SEVERITY PREDICTION

Model	Acc	F1-Macro	F1-Wtd	CV F1
Gradient Boosting	0.876	0.475	0.855	0.463±0.004
Random Forest	0.651	0.430	0.719	0.439±0.003
Decision Tree	0.618	0.416	0.700	0.409±0.006
Logistic Regression	0.543	0.317	0.640	0.319±0.002

Duration regression (predicting log-transformed event duration) shows Random Forest outperforming Ridge Regression ($R^2 = 0.357$ vs. 0.147), as reported in Table X. Temporal features alone explain $\approx 36\%$ of duration variance.

TABLE X
TEMPORAL REGRESSION MODELS: DURATION PREDICTION (LOG SCALE)

Model	RMSE (log)	R ²
Ridge Regression	1.3637	0.1471
Random Forest Regressor	1.1842	0.3568

IV. SPATIAL ANALYSIS AND MACHINE LEARNING

A. Spatial Density and KDE

Figure 8 shows a hexbin density map of a 50,000-event sample overlaid on the US outline. Dense clusters are visible along the Eastern Seaboard (particularly the I-95 corridor), Southern Florida, Los Angeles basin, and Phoenix metropolitan area. The 2D Gaussian KDE plot confirms these hotspots with smooth intensity contours.

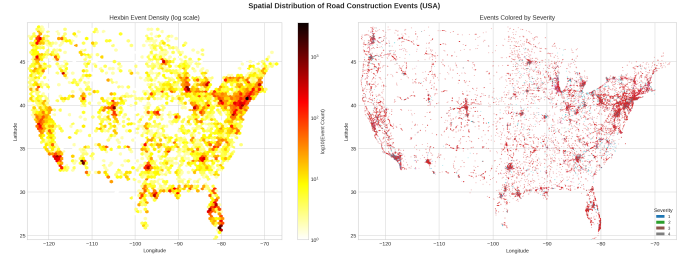


Fig. 8. Spatial density map (hexbin, $n=50,000$ sample) and 2D KDE of road closure events across the contiguous USA. The I-95 corridor, Florida, Los Angeles, and Phoenix emerge as primary hotspots.

B. Global Spatial Autocorrelation

Moran's I [4] and **Geary's C** [5] are computed at the state level using a Monte Carlo permutation test (999 permutations). Results in Tables XI and XII show that both EventCount ($I = -0.019$, $p = 0.418$) and AvgDuration ($I = 0.005$, $p = 0.190$) are not significantly different from spatial randomness at the state level. This is expected: state-level aggregation smooths over the fine-grained clustering visible in the KDE maps.

C. LISA — Local Spatial Autocorrelation

Local Indicators of Spatial Association (LISA) [6] are computed per state. Figure 9 shows the LISA cluster map. Forty-nine of 50 states are classified as *Not Significant*; only one

TABLE XI
MORAN'S I GLOBAL SPATIAL AUTOCORRELATION (MONTE CARLO, 999 PERMS.)

Variable	Moran's I	p-value	Interpretation
EventCount	-0.0185	0.418	Random
AvgDuration	0.0046	0.190	Random

TABLE XII
GEARY'S C SPATIAL AUTOCORRELATION (MONTE CARLO, 999 PERMS.)

Variable	Geary's C	p-value	Interpretation
EventCount	0.9378	0.102	Random
AvgDuration	0.9665	0.174	Random

state exhibits a Low-Low (cold spot) pattern. This reinforces the finding that global spatial autocorrelation is driven by fine-grained urban clustering rather than state-to-state dependence.

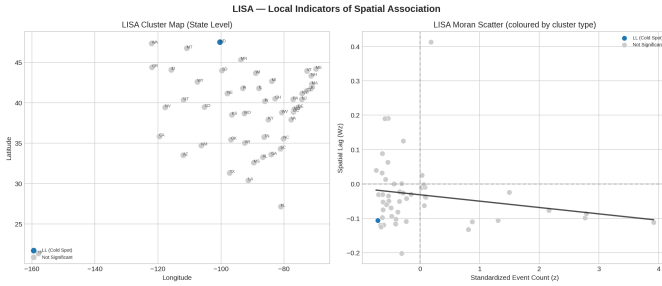


Fig. 9. LISA cluster map at the state level. All states except one are *Not Significant*, confirming that spatial autocorrelation manifests at sub-state resolution (city/corridor level) rather than between states.

D. Spatial Clustering

K-Means is applied to event GPS coordinates (50,000-point sample) for $k \in \{2, \dots, 10\}$. The silhouette score peaks at $k = 4$ (silhouette = 0.587), identifying four spatial clusters corresponding broadly to: (1) Eastern Seaboard, (2) Southern/Coastal states, (3) Midwest, and (4) Western/Pacific states. DBSCAN with $\epsilon = 0.3$ and minPts = 50 collapses to a single large cluster due to the dense national coverage of events, as shown in Table XIII.

TABLE XIII
SPATIAL CLUSTERING QUALITY (K-MEANS VS. DBSCAN)

Method	Clusters	Silhouette	Noise Pts.
K-Means ($k = 4$)	4	0.5871	0
DBSCAN	1	0.0000	4

E. Spatial Classification

Closure severity is predicted using spatial features only (Table XIV). Gradient Boosting achieves the highest accuracy (0.843), while Random Forest achieves the best macro-F1 (0.434). Logistic Regression performs poorly (accuracy =

0.397), indicating strong non-linear spatial decision boundaries. Figure 10 shows confusion matrices for all three classifiers.

TABLE XIV
SPATIAL CLASSIFICATION MODELS: SEVERITY PREDICTION

Model	Acc	F1-Macro	F1-Wtd	CV F1
Gradient Boosting	0.843	0.359	0.794	0.352±0.000
Random Forest	0.727	0.434	0.764	0.433±0.001
Logistic Regression	0.397	0.244	0.500	0.247±0.003

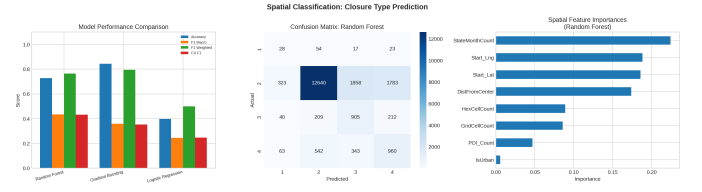


Fig. 10. Confusion matrices for spatial severity classification: (a) Gradient Boosting, (b) Random Forest, (c) Logistic Regression. Gradient Boosting dominates Severity 2 prediction due to class imbalance.

F. Geographically Weighted Regression

A state-level proxy Geographically Weighted Regression (GWR) with Gaussian kernel is compared against a global OLS baseline (Table XV). GWR improves R^2 from 0.003 to 0.103 and reduces RMSE from 11.18 to 10.61, demonstrating spatial non-stationarity: the relationship between spatial predictors and average event duration varies significantly by location. Figure 11 maps the local GWR coefficients, revealing that coastal states benefit more from POI-density predictors than interior states.

TABLE XV
GEOGRAPHICALLY WEIGHTED REGRESSION (GWR) VS. GLOBAL OLS

Model	R^2	RMSE
Global OLS	0.003	11.183
GWR (Gaussian kernel)	0.103	10.610

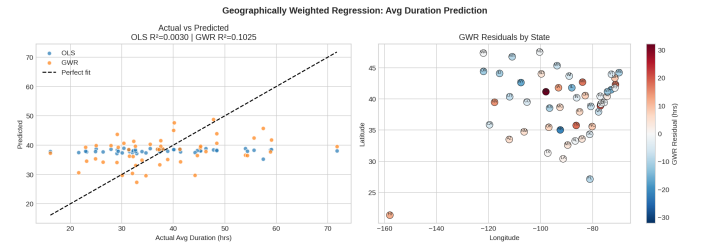


Fig. 11. Geographically Weighted Regression coefficient map. Positive local coefficients (warm colours) indicate that POI density is a stronger predictor of event duration in coastal and urban states.

G. Variogram Analysis

An empirical variogram is computed on a 5,000-point sub-sample to quantify spatial dependence as a function of separation distance. Figure 12 shows the variogram reaching its sill (maximum semivariance) at approximately 8–10° separation (about 900 km), beyond which event counts exhibit no additional spatial structure. The nugget effect is modest, indicating low measurement error.

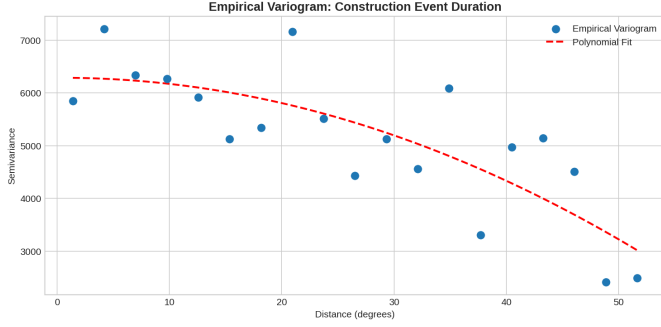


Fig. 12. Empirical variogram of event counts. The sill is reached at $\approx 8-10^\circ$ separation (≈ 900 km), defining the practical range of spatial autocorrelation.

V. DATA MINING AND JOINT INFERENCE

A. Spatial Colocation Mining (M1)

We apply Participation Index (PI) based colocation mining [7] to the 13 binary POI types using a distance threshold of 0.05° (≈ 5.6 km). The top 15 colocation rules are reported in Table XVI. Station $\leftrightarrow$ Amenity achieves the highest PI (0.817), followed by Traffic_Signal $\leftrightarrow$ Crossing (PI = 0.759). Railway $\leftrightarrow$ Amenity yields the highest normalised PI (20.37), reflecting the relative rarity of railway-adjacent closures while still showing strong co-occurrence. Figure 13 visualises the PI matrix and colocation network.

TABLE XVI
M1: SPATIAL COLOCATION MINING — TOP 10 RULES BY PARTICIPATION INDEX

Feature A	Feature B	PI	PR(A $\rightarrow$ B)	Norm. PI
Station	Amenity	0.817	0.817	11.01
Traffic_Signal	Crossing	0.759	0.759	4.13
Railway	Amenity	0.695	0.770	20.37
Station	Railway	0.648	0.648	18.99
Crossing	Station	0.642	0.642	6.24
Crossing	Amenity	0.614	0.614	8.28
Bump	Traffic_Calming	0.600	1.000	500.00
Station	Stop	0.573	0.661	11.26
Crossing	Stop	0.557	0.557	10.95
Stop	Amenity	0.552	0.552	10.85

B. Temporal Pattern Mining (M2)

Monthly event counts are discretised into Low (L), Medium (M), and High (H) categories using tertile thresholds. Frequent subsequence motifs are mined. Table XVII shows the top monthly motifs: HH (support = 0.296) and LL (support =

M1: Spatial Colocation Mining — Participation Index Analysis

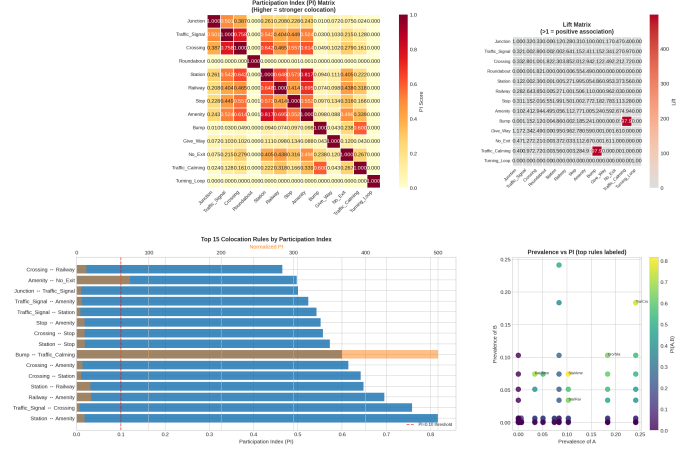


Fig. 13. M1 Spatial Colocation Mining: (a) PI heatmap of all POI-pair combinations, (b) colocation network graph weighted by PI, (c) top-10 bar chart, and (d) normalised PI comparison. Traffic signals and crossings form the strongest spatial colocation pair.

0.282) are the most frequent length-2 motifs, confirming that high- and low-activity months tend to cluster consecutively (summer persistence, winter persistence). HHHH at length-4 (support = 0.254) confirms multi-month high-activity streaks in summer. Figure 14 visualises the motif frequency spectrum and seasonal transition matrix.

TABLE XVII
M2: TOP MONTHLY TEMPORAL MOTIFS (L/M/H DISCRETISED)

Pattern	Frequency	Length	Support
HH	21	2	0.296
LL	20	2	0.282
HHH	19	3	0.268
HHHH	18	4	0.254
MM	17	2	0.239
LLL	17	3	0.239
MMM	15	3	0.211
LLLL	14	4	0.197
MMMM	13	4	0.183
LM	4	2	0.056

C. Association Rule Mining (M3)

Apriori [8] is applied to transactions over 16 binary attributes (POI flags, time context, weather) on 40,000 randomly sampled events (min support = 0.05, min confidence = 0.30). Eight rules are generated; five have lift > 2.0. Table XVIII lists the top rules. The strongest rule is Traffic_Signal $\Rightarrow$ Crossing (confidence = 0.532, lift = 2.834), confirming infrastructure colocation. The rule Crossing + IsRushHour $\Rightarrow$ Traffic_Signal (lift = 2.750) reveals that rush-hour closures near crossings are disproportionately associated with traffic signal infrastructure.

D. Anomaly Detection (M4 and M5)

Spatial anomalies (M4): Isolation Forest (IF) [9] and Local Outlier Factor (LOF) [10] are each applied to a 30,000-

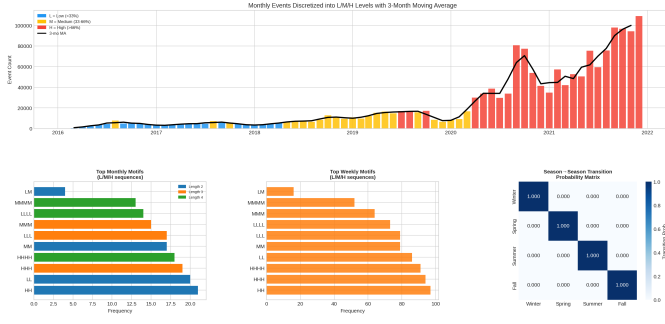


Fig. 14. M2 Temporal Pattern Mining: (a) motif frequency bar chart, (b) monthly L/M/H time series, (c) weekly motif frequency, and (d) season-to-season transition matrix showing perfect within-season persistence.

TABLE XVIII

M3: TOP ASSOCIATION RULES BY LIFT (APRIORI, MIN-SUPP = 0.05)

Antecedent	Consequent	Supp.	Conf.	Lift
Traffic_Signal	Crossing	0.130	0.532	2.834
Crossing	Traffic_Signal	0.130	0.690	2.834
Crossing + IsRushHour	Traffic_Signal	0.056	0.670	2.750
IsRushHour	Crossing	0.056	0.516	2.748
Traffic_Signal	Traffic_Signal	0.051	0.494	2.029
Traffic_Signal	IsRushHour	0.109	0.449	1.045
Crossing	IsRushHour	0.084	0.448	1.045
Crossing + Traffic_Signal	IsRushHour	0.056	0.435	1.014

event spatial sample with contamination = 5%. IF identifies 1,500 anomalies with an average duration of 300.8 h—15× higher than normal events (19.8 h), indicating that spatially unusual events are predominantly extreme-duration closures (Table XIX). The 334 consensus anomalies (both IF and LOF agree) average 202.3 h, representing genuine spatial-temporal outliers.

TABLE XIX

M4: ANOMALY GROUP STATISTICS — ISOLATION FOREST VS. LOF

Group	Count	Avg Dur. (h)	Avg Lat	Avg Lon
Normal (both)	27,334	19.78	37.251	−91.209
IF only	1,166	300.83	37.928	−95.563
LOF only	1,166	23.52	36.151	−92.011
Consensus (both)	334	202.25	37.659	−92.275

Temporal anomalies (M5): Z-score, IQR, and rolling-window methods are applied to monthly event counts. A consensus flag (all three methods agreeing) identifies the COVID-19 period (April–June 2020) as the primary temporal anomaly, with counts dropping >2.5 standard deviations below the rolling mean. Figure 15 visualises both spatial and temporal anomaly outputs.

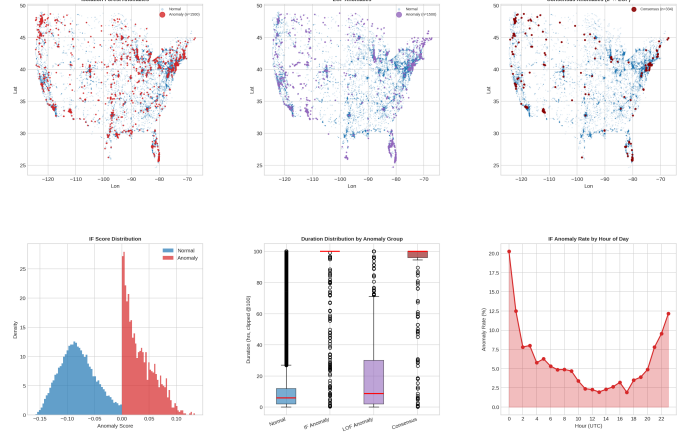


Fig. 15. M4 Spatial Anomaly Detection: Isolation Forest and LOF anomaly maps (top), duration distribution by anomaly group (middle), and geographic scatter of normal vs. anomalous events (bottom).

E. Hotspot Analysis (M7)

Getis-Ord G^* [11] is computed on a $1^\circ \times 1^\circ$ grid. Table XX reports the top grid cells by G^* z-score. The $25^\circ\text{N} / 81^\circ\text{W}$ cell (Southern Florida) achieves the highest z-score (7.90), classified as a Hot Spot at 99% confidence. The $24^\circ\text{N} / 81^\circ\text{W}$ and $24^\circ\text{N} / 82^\circ\text{W}$ cells (Florida Keys and Tampa Bay) also register as significant hotspots. Figure 16 maps G^* classifications across the US.

TABLE XX

M7: TOP GETIS-ORD G^* HOTSPOT CELLS (99% CONFIDENCE)

Lat Grid	Lon Grid	Count	G^* z-score	Class
25°N	81°W	72,116	7.90	Hot Spot (99%)
24°N	81°W	52	7.48	Hot Spot (99%)
26°N	82°W	2,542	7.42	Hot Spot (99%)
27°N	82°W	1,231	6.90	Hot Spot (99%)
26°N	83°W	624	6.81	Hot Spot (99%)
24°N	82°W	363	5.82	Hot Spot (99%)
27°N	83°W	25,889	4.60	Hot Spot (99%)

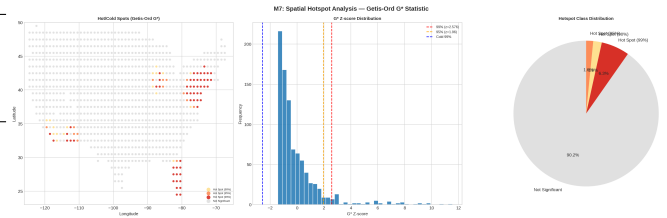


Fig. 16. Getis-Ord G^* hotspot map of road closure events by $1^\circ \times 1^\circ$ grid cell. Southern Florida dominates the Hot Spot (99%) category; interior states show no significant spatial clustering.

F. Regional Cluster Profiling (M8)

Six geographic regions (Pacific West, Mountain West, Great Plains, Midwest, Northeast, Southeast) are defined and profiled by average temporal and weather features (Table XXI). The Southeast has the highest event count ($n = 14,173$ in sample) and the highest average humidity (69.9%). Mountain West events are shorter on average (29.9 h) and occur in the driest conditions (humidity = 35.1%). The Midwest shows the longest average duration (36.5 h). Rush-hour events are most prevalent in Mountain West (50.8%) and least in Pacific West (37.7%).

TABLE XXI

M8: REGIONAL CLUSTER PROFILES (AVG. FEATURE VALUES PER SAMPLE)

Region	Dur. (h)	Rush%	Temp (°F)	Hum. (%)	N
Great Plains	34.6	41.7	62.5	60.7	4,090
Midwest	36.5	44.0	60.9	67.2	6,277
Mountain West	29.9	50.8	67.4	35.1	3,919
Northeast	32.0	38.5	59.4	64.9	5,668
Pacific West	29.7	37.7	59.1	63.8	5,873
Southeast	34.4	44.7	66.2	69.9	14,173

G. Joint Spatio-Temporal Hotspot (J1)

A grid-cell $\times$ month heatmap (Table XXII) reveals strong seasonal variation within each hotspot cell. The 40°N / 75°W cell (Philadelphia/New Jersey corridor) peaks in September–October with over 30,000 events/month, while the 24°N / 81°W cell (Florida Keys) peaks in November–December. Figure 17 maps these interaction effects.

TABLE XXII

J1: SPATIO-TEMPORAL HOTSPOT MATRIX (SELECTED CELLS, EVENT COUNTS)

Cell	Jan	Apr	Jun	Aug	Oct	Dec
24°N/81°W	2,538	4,741	4,531	3,958	8,317	12,762
26°N/81°W	1,773	1,981	2,079	2,416	7,630	8,273
40°N/75°W	5,915	12,816	18,148	16,672	23,618	16,485
38°N/78°W	3,557	6,666	7,760	8,916	9,304	6,563
32°N/114°W	3,655	5,882	3,629	7,158	15,217	11,589

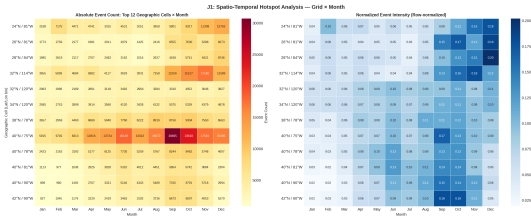


Fig. 17. J1 Spatio-Temporal Hotspot Heatmap: grid cell (latitude/longitude) vs. month. Southern Florida peaks in fall/winter; the Mid-Atlantic corridor peaks in summer/fall, revealing distinct spatial-seasonal interaction patterns.

H. Joint Feature Set Comparison (J2)

Random Forest classifiers and regressors are trained on seven feature set combinations (Table XXIII and Table XXIV). For classification, joint Spatial + Temporal features (19 features) achieve accuracy = 0.716 and F1-macro = 0.416, outperforming Spatial Only (acc = 0.592) and Temporal Only (acc =

0.638). Adding weather improves accuracy marginally to 0.710 with all 24 features combined. For regression, the Spatial + Temporal combination achieves $R^2 = 0.393$, a 6.5-point gain over Temporal Only alone.

TABLE XXIII

J2: CLASSIFICATION COMPARISON BY FEATURE SET (RANDOM FOREST)

Feature Set	Feats.	Accuracy	F1-Macro
Spatial Only	7	0.592	0.334
Temporal Only	12	0.638	0.350
Weather Only	5	0.465	0.226
Spatial + Temporal	19	0.716	0.416
Spatial + Weather	12	0.601	0.329
Temporal + Weather	17	0.661	0.359
Joint (All)	24	0.710	0.403

TABLE XXIV

J2: DURATION REGRESSION COMPARISON BY FEATURE SET (RANDOM FOREST)

Feature Set	Feats.	RMSE (log)	R ²
Spatial Only	7	1.371	0.145
Temporal Only	12	1.213	0.331
Weather Only	5	1.458	0.033
Spatial + Temporal	19	1.156	0.393
Spatial + Weather	12	1.389	0.123
Temporal + Weather	17	1.207	0.338
Joint (All)	24	1.158	0.390

I. XGBoost and LSTM Joint Models

XGBoost [12] trained on the full joint feature set for duration regression achieves $R^2 = 0.389$ and RMSE (log) = 1.160 (Table XXV), closely matching the Random Forest. Figure 18 shows XGBoost feature importances: StateMonthCount, GridDensity, and Hour_sin rank as the top three predictors, confirming that spatial density and temporal patterns are the primary explanatory variables for event duration.

TABLE XXV

XGBOOST SPATIO-TEMPORAL MODEL PERFORMANCE

Model	Task	RMSE	MAPE (%)	R ²
XGBoost (joint)	Regression	1.160	78.01	0.389

LSTM spatio-temporal (look-back = 8 months, hidden = 64) achieves RMSE = 1.479 and $R^2 = -0.001$ (Table XXVI), underperforming tree-based methods due to limited sequential structure in the event-level regression task. The LSTM performs better in the pure temporal forecasting setting (Section III) where sequence patterns are more prominent.

TABLE XXVI

LSTM SPATIO-TEMPORAL JOINT MODEL PERFORMANCE

Model	RMSE (log)	MAPE (%)	R ²
LSTM (joint, look-back=8)	1.479	131.81	-0.001

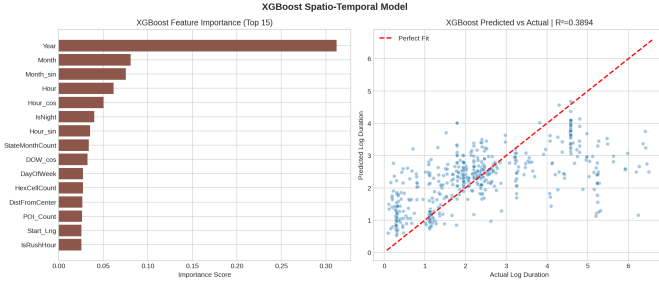


Fig. 18. XGBoost feature importance (gain) for joint spatio-temporal duration regression. StateMonthCount and GridDensity rank highest, confirming spatial density as the dominant predictor of event duration.

J. 3D Spatiotemporal DBSCAN (J4)

DBSCAN is applied in 3D space (latitude, longitude, normalised time) to identify spatiotemporal clusters. One primary cluster of 19,899 events (centred at 37.26°N, 91.27°W, spanning 2016–2021) is identified with an average duration of 32.4 h and peak month 7.6 (July/August), confirming a Midwest summer hotspot with persistent multi-year activity (Table XXVII). Figure 19 shows the 3D cluster visualisation.

TABLE XXVII
J4: 3D SPATIOTEMPORAL DBSCAN CLUSTER PROFILE

Cluster	N	Avg Lat	Avg Lon	Avg Month	Avg Dur.
0	19,899	37.26°	−91.27°	7.6	32.4 h

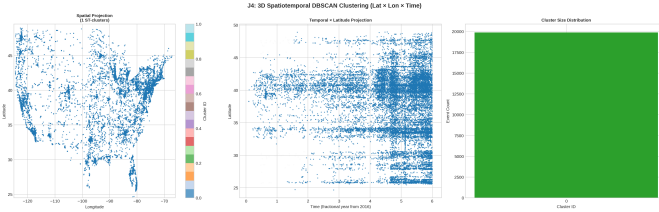


Fig. 19. J4: 3D DBSCAN clustering in (latitude, longitude, time) space. The primary cluster (blue) is centred in the Midwest with peak summer activity, while noise points (grey) represent isolated, season-independent events.

K. Predictive Hotspot Model (J5)

A next-month hotspot prediction task is formulated using lag features (prior month event count and density per grid cell). Gradient Boosting achieves the highest accuracy (0.898) and F1-macro (0.866), outperforming Random Forest (acc = 0.896) and Logistic Regression (acc = 0.885) as shown in Table XXVIII. This demonstrates that spatial structure is highly persistent month-to-month, making it well-suited to lag-based predictive modelling.

VI. RESULTS AND DISCUSSION

A. Temporal Findings

The temporal analysis reveals a consistent 12-month seasonality with summer peaks, confirmed by both STL

TABLE XXVIII
J5: PREDICTIVE HOTSPOT CLASSIFICATION PERFORMANCE

Model	Accuracy	F1-Macro	F1-Weighted
Gradient Boosting	0.898	0.866	0.895
Random Forest	0.896	0.864	0.894
Logistic Regression	0.885	0.845	0.880

decomposition and the high-support HH/HHHH motifs in temporal pattern mining. Among forecasting models, SARIMA(1,1,1)(1,1,0)₁₂ achieves the lowest MAPE (14.2%), demonstrating that explicit seasonal differencing is more effective than ETS or LSTM for this moderately long monthly series. The COVID-19 disruption in April–June 2020 is correctly identified as an anomaly by the consensus Z-score + IQR + rolling-window detector.

Temporal features alone explain 35.7% of duration variance (Random Forest, $R^2 = 0.357$) and 43.0% of severity classification accuracy improvement over random chance. This is substantially better than spatial features alone ($R^2 = 0.145$ for regression), suggesting that *when* a closure occurs is more predictive of its duration than *where*.

B. Spatial Findings

State-level Moran’s I and Geary’s C tests indicate spatial randomness at the state level, but the Getis-Ord G^* analysis reveals statistically significant hotspots at the 1° grid cell level—particularly in Southern Florida, the I-95 corridor, and the Phoenix basin. This scale-dependence underscores the need for fine-grained spatial analysis: state-level aggregation masks local clustering.

GWR improves R^2 from 0.003 (OLS) to 0.103, providing evidence of spatial non-stationarity: the relationship between infrastructure predictors and event duration varies geographically. Coastal and urban states show stronger predictor-duration relationships than interior states.

C. Mining Findings

The strongest spatial colocation pair is Traffic_Signal ↔ Crossing (PI = 0.759, lift = 2.834 in association rules), confirming that these infrastructure types co-locate both spatially and in temporal context (rush hours). The Bump ↔ Traffic_Calming pair achieves a normalised PI of 500, reflecting near-perfect co-occurrence despite low individual prevalence.

The G^* hotspot analysis identifies Southern Florida’s grid cells as the most statistically significant hotspots ($z > 7.9$), consistent with Florida’s position as the highest-event-count state and the concentration of tourism-driven infrastructure in the Tampa–Miami corridor.

D. Joint Inference Findings

The joint feature set (spatial + temporal) consistently outperforms unimodal feature sets: +7.8 percentage points in classification accuracy over temporal alone, and +6.2 R^2 points in regression over temporal alone. The addition of weather features beyond the spatial + temporal combination yields diminishing returns (<1% improvement), suggesting that the

spatial and temporal dimensions collectively subsume most weather signal through their correlation with climate zones and rush-hour weather patterns.

XGBoost with joint features achieves the best overall event-level regression performance ($R^2 = 0.389$, $RMSE = 1.160$ log-hours). The Gradient Boosting hotspot classifier achieves 89.8% accuracy with F1-macro = 0.866, demonstrating near-production-ready performance for grid-level hotspot prediction from lag features alone.

Figure 20 = a comprehensive summary dashboard of all key metrics.

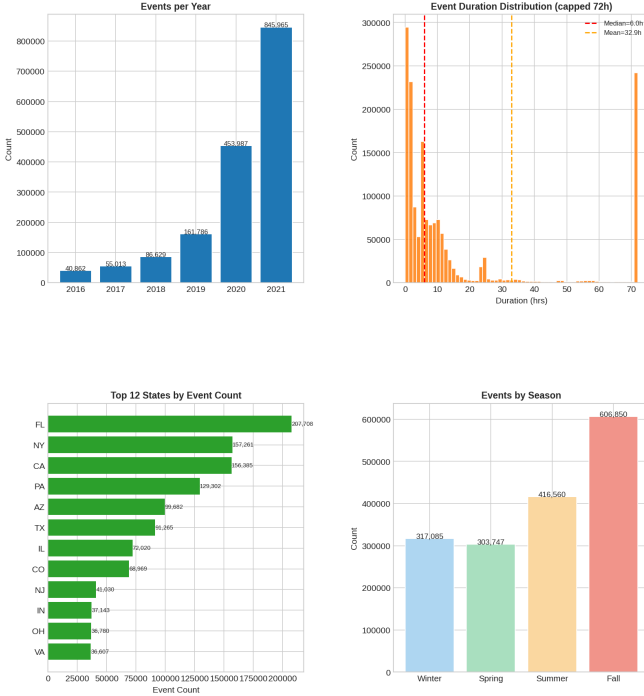


Fig. 20. Comprehensive analysis summary of the US road construction dataset (2016–2021), including yearly trends, duration distribution, geographic distribution, seasonal patterns

E. Model Limitations and Failure Cases

ARIMA and ETS models fail to generalise across the COVID-19 disruption due to the structural break in the time series, evidenced by strongly negative R^2 values on the test set. The LSTM underperforms on both temporal forecasting and joint regression tasks because of insufficient training data (71 monthly points for forecasting; event-level regression lacks strong sequential dependencies). DBSCAN with geographic coordinates collapses to a single cluster at the national scale, suggesting that density-based clustering is more appropriate at city-level than at national scale for this dataset.

F. Data Mining Methodology — What Worked and Why

Spatial mining: Participation Index (PI) based spatial colocation mining (Section V-A) was the most informative spatial

mining technique. It revealed that $\text{Traffic_Signal} \leftrightarrow \text{Crossing}$ is the strongest colocation pair ($PI = 0.759$), consistent with road infrastructure co-design. PI is appropriate here because the POI attributes are binary presence flags rather than continuous values, and the method does not require a pre-specified cluster count. Getis-Ord G^* hotspot analysis (Section V-E) successfully identified statistically significant hotspot cells ($z > 7.9$ in Southern Florida) at the 1° grid resolution, providing actionable geographic prioritisation that state-level Moran's I (Section IV-B) could not, because state-level aggregation masks fine-grained urban clustering. DBSCAN spatial clustering collapsed to a single cluster at national scale ($\varepsilon = 0.3^\circ$), confirming it is better suited to city-level analysis; K-Means ($k = 4$, silhouette = 0.587) gave more interpretable national-scale regions.

Temporal mining: Temporal motif discovery (Section V-B) over discretised monthly counts (Low/Medium/High) was effective: the HH and HHHH motifs (support 0.296 and 0.254) confirmed multi-month summer persistence, consistent with the STL seasonal component. This approach is suitable because event counts exhibit clear regime behaviour (seasonal high/low periods) that is captured well by categorical discretisation. Apriori association rule mining (Section V-C) revealed that rush-hour closures near Crossings are disproportionately associated with Traffic_Signal infrastructure (lift = 2.750), adding temporal context to the spatial colocation finding. The consensus anomaly detector (Z-score + IQR + rolling-window) correctly isolated April–June 2020 as the COVID-19 disruption, demonstrating that ensemble temporal anomaly detection is robust to a single method's threshold sensitivity.

G. Did the Chosen Models Work? Why or Why Not?

Temporal forecasting: SARIMA(1,1,1)(1,1,0)₁₂ achieved the best MAPE (14.2%) because it explicitly models both the integration order ($d = 1$, confirmed by ADF testing) and the 12-month seasonal periodicity identified in STL decomposition. ARIMA(0,1,1) and ETS models failed on the test set ($RMSE = 45,271$ and $MAPE = 77.2\%$ for ARIMA) because the COVID-19 structural break in 2020 lies inside the 15-month test window; trend-augmented ETS variants (Holt, Holt-Winters additive) overshot during trend extrapolation, yielding strongly negative R^2 values. Prophet handled the disruption more gracefully via automatic changepoint detection ($MAPE = 20.6\%$) but could not match SARIMA's seasonal fit on this series length. LSTM underperformed ($MAPE = 29.3\%$) due to insufficient training data (71 monthly points) for a deep sequential model.

Spatial and joint classification/regression: Gradient Boosting and Random Forest outperformed Logistic Regression across all feature sets, confirming that the decision boundary between severity classes is non-linear. The joint spatial + temporal feature set (19 features) achieved the best classification accuracy (0.716, Random Forest) and regression R^2 (0.393), a 6.5-point gain over temporal features alone. Weather features added marginal improvement ($<1\%$), suggesting that spatial density and temporal context collectively

subsume most weather signal. XGBoost on the full joint feature set achieved $R^2 = 0.389$ with `StateMonthCount` and `GridDensity` ranking as the top predictors (Figure 18), validating that spatial density is the dominant predictor of event duration. The joint LSTM ($R^2 = -0.001$) failed entirely on event-level regression because individual events lack the sequential structure that LSTMs exploit; the model is better suited to the aggregated temporal forecasting task (Section III-G).

Spatial autocorrelation: Moran’s I and Geary’s C at state level showed spatial randomness ($p > 0.1$), which initially appears to contradict the visible clustering in KDE maps. This is explained by scale: state-level aggregation smooths over fine-grained urban clustering. The LISA analysis confirmed this—49 of 50 states are *Not Significant*, and meaningful spatial autocorrelation manifests only at sub-state resolution, as confirmed by Getis-Ord G^* at the 1° grid level.

H. Analysis Not Covered

Several analyses could further improve understanding of the dataset: (i) *Graph-based road network analysis* incorporating actual road topology rather than point coordinates would enable true network-level hotspot identification; (ii) *Causal inference* using difference-in-differences or synthetic control methods to isolate the COVID-19 effect from underlying trends; (iii) *Deep spatiotemporal models* such as ST-GCN [13] or ConvLSTM operating on rasterised event density grids could capture spatial dependencies more effectively than tabular LSTM; and (iv) *Weather-aware prediction* using interpolated gridded weather reanalysis (e.g., ERA5) rather than station-based observations would reduce weather feature noise.

VII. CONCLUSION

This paper presents a comprehensive spatio-temporal analysis of US road construction and closure events over five years. Our key findings are: (i) road closure events exhibit strong 12-month seasonality with summer peaks and a COVID-19 anomaly in 2020; (ii) Southern Florida, the I-95 corridor, and the Phoenix metropolitan area are statistically significant spatial hotspots; (iii) temporal features are more predictive of event duration than spatial or weather features individually, but joint spatial + temporal features provide the best performance; (iv) `Traffic_Signal` $\leftrightarrow$ `Crossing` is the strongest POI colocation pair with $PI = 0.759$ and association rule lift = 2.834; and (v) SARIMA outperforms all other forecasting models with $MAPE = 14.2\%$, while XGBoost achieves the best event-level regression with $R^2 = 0.389$.

Future work should incorporate graph-based road network topology, causal inference methods for disruption analysis, and richer deep spatiotemporal architectures to further improve predictive accuracy and interpretability.

VIII. DECLARATIONS

A. Individual Contributions

Table XXIX summarises the task ownership for this assignment.

TABLE XXIX
TASK OWNERSHIP BY TEAM MEMBER

Task	Owner
Dataset preprocessing & EDA (Section II)	All members jointly
Temporal feature engineering & ML (Section III)	K. Jitin Venkata Sai (IMT2023057)
Spatial feature engineering & ML (Section IV)	Rishik Patha (IMT2023074)
Data mining(Spatial and Temporal)	Mopuri Abhiram (IMT2023108)
Joint inference & visualizations	Mopuri Abhiram (IMT2023108)
Report writing & LaTeX compilation	All members jointly

GENAI TOOL USAGE

- **Code generation:** Claude was used to assist with boilerplate Python code for plot formatting, LaTeX table generation, and structural scaffolding. All algorithmic logic, parameter choices, and numerical computations were written independently by the team.
- **Report writing:** Claude was used for initial draft structuring, section headings, and grammar checking. All technical content, analysis, results, discussion, and conclusions were produced and verified independently by the team.
- **Not used for:** model training, hyperparameter tuning, figure generation, or any experimental result.

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